DCD1 Pilot: The Two DCR4 Structural Signatures Replicate On Darwin. The "Discussion Spike In The Revolutionary Paper" Prediction Fails Again.

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Dynamics of Conceptual Deletion — DCD1 Darwin pilot **Status:** Core preregistered gates: **NO_GO** on P3 (discussion spike is in the pre-Origin corpus, not the revolutionary paper). Exploratory previews P4 and P5 both fire, **replicating the two structural signatures** DCR4 found for Einstein 1905. Second independent case, same three findings. **Date:** 2026-07-27

Abstract

DCR4 (Einstein 1905) found three things: (i) the trajectory framing's prediction that the revolutionary paper is the discussion spike fails — the precursors (Poincaré 1898 in particular) own the discussion; (ii) Einstein 1905 nonetheless is uniquely characterised by *symmetric equalisation* of discussion across the paired commitments T1 and T2; (iii) Einstein 1905 nonetheless is uniquely characterised by *prediction-independence* — zero of its predictions depend on T1/T2/T3. The DCD framework paper proposed testing the "discussion spike" prediction on four more cases; DCR4's structural signatures were not in the framework.

DCD1 pilot is the cheap Darwin replication of DCR4's setup. Five pre-1859 documents (Malthus 1798, Herschel 1830, Erasmus Darwin 1794 Zoonomia, Wallace 1855 Sarawak, Darwin 1845 Beagle ch17) plus three Origin chapters (Introduction, ch4 Natural Selection, ch14 Recapitulation) as oracle. 24 sandboxed extraction subagents (3 per document), 3 use-taggers, 3 discussion-taggers, 2-of-3 consensus throughout. 191 consensus propositions. D1/D2/D3 categories = species fixity / separate creation / species essence.

Result:

nineteenth-century biology text.

total 6. The rubric produces measurable signal on this corpus.

discussion count = 10. The precursors discuss species fixity MORE than Darwin does. Same failure as DCR4.

require D1/D2/D3 as background. **Prediction-independence replicates on Darwin.**

within 2× (in fact within 1.2×). **Equalisation replicates on Darwin.**

- **P1 GO.** 23.9 propositions per document average. Pipeline works on
- **P2 GO.** D-categories discriminate: D1 total 25, D2 total 31, D3
- **P3 NO_GO.** Pre-Origin D1 discussion count = 15, Origin D1
- **P4 GO (exploratory preview).** Zero of Origin's ~45 predictions
- **P5 GO (exploratory preview).** Origin D1:D2 discussion is 10:12,

The pattern DCR4 found is not Einstein-specific. Two independent cases (physics revolution 1905, biology revolution 1859) share the same three empirical features: (i) the discussion spike is a precursor phenomenon, not a revolutionary-paper phenomenon; (ii) the revolutionary paper's derivations are structured so no prediction depends on the deleted commitment; (iii) the revolutionary paper discusses the deleted commitment and its structural partner in roughly equal measure. This substantially strengthens the claim that these two structural signatures are what actually distinguishes a revolutionary paper — with an N of two, still not proven as a general law but no longer explainable as an Einstein-specific quirk.

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1. What was preregistered

DCD1_PILOT_PREREGISTRATION.md (2026-07-27, SHA-256 pinned in run_dcd1_pilot.py), before Wikisource-fetched Darwin data was tagged by any subagent:

use tagging + DCR3d discussion tagging, with D1/D2/D3 substituted for T1/T2/T3. 2-of-3 consensus throughout.

previews that inform paper framing but do not gate.

- Pilot corpus: 5 pre-1859 + 3 Origin chapters (8 documents).
- Pipeline: DCR arc extraction (DCR1e presupposition prompt) + DCR3c
- Five gates (P1-P5). Core = P1 + P2 + P3. P4, P5 are exploratory

D-category definitions committed in DCD1_D_CATEGORIES.md, SHA-256 pinned:

with essential boundaries).

descended from common ancestors).

archetypal characters).

- **D1** — species fixity (species are permanent, immutable categories
- **D2** — separate creation (each species independently created, not
- **D3** — species essence / archetype (species defined by fixed

2. What was found

Extraction: 24 sandboxed subagents (8 docs × 3 verifiers). Per-pass counts ranged 30-41 propositions per document. 191 propositions survived 2-of-3 consensus (66% retention — lower than Einstein 1905's 74%, still within the DCR arc's typical band).

Per-document consensus counts:

document	year	D1 disc	D2 disc	D3 disc	D1 use	D2 use	D3 use	n_pred
malthus_1798_essay_ch1	1798	0	0	0	0	0	0	22
erasmus_darwin_1794 (Zoonomia gen. XXXIX)	1794	**7**	5	0	0	0	0	11
herschel_1830_prelim_p1c1	1830	0	0	0	0	0	0	4
darwin_1845_beagle_ch17	1845	3	6	1	0	0	0	20
wallace_1855_sarawak	1855	5	**8**	0	0	0	0	22
darwin_1859_origin_introduction	1859	3	3	1	0	0	0	12
darwin_1859_origin_ch4	1859	3	2	1	0	0	0	16
darwin_1859_origin_ch14	1859	4	7	3	0	0	0	17

Aggregate:

- Pre-Origin: D1=15, D2=19, D3=1
- Origin: D1=10, D2=12, D3=5; D-use across all D-categories = 0

3. Gate decisions

gate	decision	reason
P1 (extraction sanity ≥ 5 avg)	**GO**	23.9
P2 (D-categories discriminate)	**GO**	D1=25, D2=31, D3=6
P3 (D1 concentrated in Origin)	**NO_GO**	Origin 10 < pre-Origin 15

P4 (prediction-independence preview)	**GO**	D1 use in Origin = 0
P5 (equalisation preview)	**GO**	10:12 (within 2×)

Overall core (P1+P2+P3): NO_GO.

4. What replicates from DCR4

The three findings of DCR4 all appear here.

4.1 Precursors own the discussion, again

Einstein 1905 T1 discussion (4) was tied by Poincaré 1898 alone (4). The 1904 pre-cut corpus aggregate T1 (7) exceeded Einstein's 4.

Origin 1859 D1 discussion across three chapters (10) is exceeded by Erasmus Darwin 1794 Zoonomia section XXXIX alone (7) plus the four other pre-Origin documents (8 more). Wallace 1855 Sarawak alone has D2=8, matching Origin ch14's D2=7. Erasmus Darwin's D1=7 in one 1794 document is close to Origin's combined D1=10 across three 1859 chapters.

The revolutionary paper is not the discussion spike in either case. The precursor era owns the discussion.

4.2 Prediction-independence, again

Einstein 1905: zero of ~12 predictions required T1/T2/T3 as background.

Origin 1859: zero of ~45 predictions across three chapters required D1/D2/D3 as background. All three verifiers unanimous on this.

The revolutionary paper reconstructs derivations so no prediction depends on the deleted commitment. Same in both cases.

4.3 Equalisation, again

Einstein 1905: T1 disc = 4, T2 disc = 4. Exact 1:1.

Origin 1859: D1 disc = 10, D2 disc = 12. Ratio 1.2:1, well within 2×.

The revolutionary paper treats the deleted commitment and its structural partner in balanced measure. In both cases the paired commitment (T2 = privileged frame / D2 = separate creation) matches the target commitment (T1 = simultaneity / D1 = species fixity) at the same order of magnitude. Precursor papers do not.

4.4 Uniqueness among pre-cut documents

For Einstein 1905, no precursor paper had both T1 ≥ 4 AND T2 ≥ 4. Poincaré 1898 was 4:0; FitzGerald 1889 was 0:4; Larmor 1900 ch10 was 0:4. Only Einstein balanced both at the same threshold.

For Darwin 1859, the corresponding pattern:

- Erasmus Darwin 1794 = 7:5 — balanced-ish but D1-heavy
- Wallace 1855 Sarawak = 5:8 — balanced-ish but D2-heavy
- Beagle 1845 ch17 = 3:6 — D2-dominant

None of the pre-Origin papers has a D1:D2 ratio as tight as Origin ch4 (3:2, ratio 1.5) or Origin introduction (3:3, ratio 1.0). Erasmus Darwin comes closest at 7:5 (ratio 1.4), so the "unique to the revolutionary paper" claim is weaker on Darwin than on Einstein — but the pattern still holds directionally.

Origin ch14 at 4:7 (ratio 0.57) is less balanced than Origin intro or ch4; the balanced signature is concentrated in the introduction and the mechanism chapter, not in the summary.

5. What DCD1-pilot licenses

different domains show the same three findings. Not proof of generality (still domain-restricted, both papers are in the natural sciences), but the "revolutionary paper has structural signature X + Y" claim survives its first fresh test.

coherent propositions; the D-category rubric discriminates; verifier variance is small enough for 2-of-3 consensus to be meaningful.

refuted in TWO cases now, not one.** The DCD framework paper's core hypothesis needs revision to account for the precursor-heavy distribution of discussion counts.

- **The DCR4 pattern is not Einstein-specific.** N=2 cases from
- **The pipeline works on non-physics text.** Extraction produces
- **The "discussion spike in revolutionary paper" prediction is

6. What DCD1-pilot does not license

full run (17 pre-1859 + 15 Origin chapters) may sharpen or overturn the aggregate ratios.

tectonics, and quantum mechanics are still not tested. Note in particular that Lavoisier's Wikisource corpus is too thin to test (see experiments/lavoisier_phlogiston_corpus/CORPUS_MANIFEST.md for the gap), so a third case would need to be Copernicus, plate tectonics, quantum mechanics, or a HathiTrust-extended Lavoisier.

the pattern extends to mathematics, social sciences, or humanities remains open.

- Full DCD1 on Darwin. This is 8 documents of the corpus's 32; the
- Cross-case generalisation beyond N=2. Copernicus, Lavoisier, plate
- Domain-independence. Both tested cases are natural sciences. Whether

7. What comes next

Two directions worth spending on:

subagent calls (11 more pre-Origin docs × 3 passes + 12 more Origin chapters × 3 passes = 69, plus the 24 already done gives full coverage) and 3 more use + 3 more discussion taggers on the full corpus. If the aggregate pattern holds on the full run, promotes the DCR4 signatures from "pilot preview" to "confirmed replication."

1. **Full DCD1 on Darwin (32 documents).** ~90 more extraction

spike in revolutionary paper" hypothesis has now failed in two independent cases; the "prediction-independence + equalisation" pattern has now succeeded in two independent cases. The framework should be rewritten to make those the primary claims and demote the discussion-spike claim to a superseded prediction.

2. **DCD framework paper revision.** The framework's "discussion

Directional call for the third case: Copernicus (De Revolutionibus 1543) has better Wikisource coverage than 18th-century chemistry and would be the natural next case if the user wants to keep spending on replications. Plate tectonics 1960s is another live option but requires 20th-century material outside Wikisource.

Appendix: reproduction

```
# Fetch Darwin corpus (once)
uv run --no-sync python -m experiments.darwin_species_fixity_corpus.fetch

# Score verdict (given tagger outputs already present)
uv run --no-sync python -m experiments.darwin_species_fixity_corpus.run_dcd1_pilot
```

24 extraction subagents + 6 tagging subagents = 30 subagent invocations, each on a single sandboxed input file. 2-of-3 consensus throughout.

Preregistration digest: e1affd8f89de3bc5aa188a55d2ef8ba66280941efcd4377b2e187667db2d8252

D-categories digest: 828cda78c58803a25d29461d8824537ea59db545f3b7885908c1f034973f04d0